

Deep Patel

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Professional Summary

Computer Science graduate currently pursuing an M.Sc. in Artificial Intelligence at BTU Cottbus (4th Semester). Hands-on experience developing autonomous agent architectures, text-to-code pipelines, and multimodal data applications using the GenAI SDK, Groq API, and Streamlit. Proficient in implementing data preprocessing frameworks, statistical feature evaluation, and machine learning lifecycle deployments. Seeking a student level engineering placement, internship, or a 6-month Master's thesis partnership to optimize data and automation pipelines.

Technical Skills

- **Languages:** Python, SQL, HTML5, CSS3, JavaScript (ES6+), JSON
- **Libraries & Frameworks:** Scikit-Learn, XGBoost, Google GenAI SDK, Pydantic, NLTK, NumPy, Pandas, Seaborn, Matplotlib
- **Developer Tools:** Git, GitLab, GitHub REST API, Groq Cloud, VS Code, Overleaf (L^AT_EX)
- **Core Methodologies:** Reasoning-and-Acting (ReAct) Agent Architecture, Multimodal Data Pipelines, ETL Cleaning Loops, Feature Binarization, Statistical Data Preprocessing, Continuous Git Deployment

Education

Brandenburg University of Technology (BTU)

Cottbus, Germany

M.Sc. in Artificial Intelligence

2024 – Present

- **Academic Focus:** Automated code verification, multi-agent loops, reasoning-and-acting (ReAct) frameworks, and neural network engineering.
- **Thesis Track:** Open to industrial thesis partnerships or engineering team placements (14-20h/week).

Parul University

Gujarat, India

B.Tech. in Computer Science and Engineering

2019 – 2023

- **Technical Foundations:** Gained core competencies in application logic, object-oriented concepts, system normalization, software testing parameters, and comprehensive relational database integration labs.

AI & Machine Learning Developer Portfolio (Master Repository)

Personal Developer Web Portfolio

JavaScript, GitHub API, HTML5, CSS3

Problem Solved: Streamlines personal branding by dynamically syncing live software development updates and project repos without manual website maintenance.

- Designed and deployed a responsive developer portfolio website tracking active software projects.
- Integrated the GitHub REST API to asynchronously fetch, parse, and dynamically display live repository folders and update histories directly on the UI layout.
- Programmed a secure, functional frontend asynchronous contact form validating user submission fields.

Python ReAct Automation Agent Framework

Python, Streamlit, Groq API, Regex

Problem Solved: Automates complex, dynamic, tool-based reasoning tasks without relying on external agent frameworks, optimizing token usage and workflow tracking.

- Programmed a library-free, zero-dependency implementation of the Reasoning and Acting (ReAct) agent loop pattern using pure Python logic.
- Deployed an interactive web dashboard via Streamlit hooked directly to the code repository to stream real-time thought chains, tool execution blocks, and token counts.
- Leveraged regular expressions to isolate model text sequences into valid tool instructions and functional string outputs.

AI Resume Analyzer Pro

Python, Streamlit, Google GenAI SDK, Pydantic

Problem Solved: Resolves structural gaps and parsing layout vulnerabilities in raw unstructured text resumes relative to modern automated corporate technical filters.

- Developed a multi-widget web dashboard using Streamlit to handle file uploads and parse unstructured raw text.
- Integrated an LLM API client via the google-genai SDK using Pydantic schemas to enforce structured JSON outputs on gemini-3.1-flash-lite.
- Automated deployment pipelines through a direct GitHub repository connection for seamless version delivery.

AI YouTube Video Analyzer

Python, Streamlit, YouTube Transcript API, JSON

Problem Solved: Eliminates long video evaluation times by programmatically extracting audio text tracks and structuring massive raw multi-column scripts into clean, readable formats.

- Built a standalone analytics pipeline parsing video data structures to generate concise topical abstracts and action items.

- Configured an automated data connection layer fetching transcriptions using standard structural API calls via the youtube-transcript-api.
- Developed clean rendering tables inside the UI framework to format multi-column summary text blocks dynamically.

California House Price Prediction Model

XGBoost Regressor, Seaborn, Pandas

Problem Solved: Protects pricing models from geometric prediction distortions caused by dense feature collinearity across spatial matrix vectors.

- Imported the California housing database via Scikit-Learn to build a predictive price forecasting matrix consisting of 20,640 instances and 8 predictive attributes.
- Generated a Seaborn correlation heatmap to visualize linear relationships and multi-collinearity between dataset features.
- Evaluated target test predictions across split data frames achieving a highly accurate R^2 score of 0.943 on training data and 0.833 on test data, alongside a Mean Absolute Error (MAE) of 0.193 on training.

Fake News Text Processing Pipeline

NLTK, TfidfTransformer, Logistic Regression

Problem Solved: Resolves text corpus variance and tracking noise from alphanumeric strings to construct high-accuracy news classification indicators.

- Cleaned an extensive textual dataset of 72,134 news articles using Python's regular expressions library to strip out trailing characters and non-alphabet markers.
- Leveraged the NLTK library to extract root definitions via a PorterStemmer framework while dropping stop-word tokens.
- Transformed text weights using a TfidfTransformer layer to build a Logistic Regression model achieving 92.7% accuracy on training data and 90.3% accuracy on test data.

Automated Loan Status Prediction Engine

Support Vector Machines, Pandas

Problem Solved: Eliminates transactional screening errors by systematically cleaning sparse financial metrics and structural row variance.

- Processed an operational loan application dataset of 614 records by dropping missing rows (resulting in a clean 480-record dataset) and encoding categorical strings into binary arrays.
- Replaced conflicting alphanumeric characters across Pandas data blocks to map uniform column scaling patterns.
- Split historical inputs using a stratified framework to fit a baseline linear Support Vector Machine layout, yielding an 83.3% accuracy score on test predictions.

Sonar-Array Diagnostics Engine

Scikit-Learn, Logistic Regression, Pandas, NumPy

Problem Solved: Automates binary geological categorization (Mine vs. Rock) over dense frequency spectrum matrix variables, eliminating hand-coded signal filtering.

- Processed a 60-feature raw sonar frequency spectrum dataset containing 208 instances to map signal variables against binary geological targets.
- Engineered stratified framework boundaries to isolate feature sets and mitigate model target assignment skew.
- Trained a robust Logistic Regression classification model, securing an 83.4% accuracy matrix on training data and 76.1% accuracy across split test iterations.

PIMA Diabetes Diagnostic Pipeline

Streamlit, StandardScaler, Support Vector Machines

Problem Solved: Eradicates predictive structural skew caused by conflicting metric scales during the processing of real-time clinical indicator feeds.

- Deployed a live, user-facing linear Support Vector Classifier (SVC) web application evaluating 768 patient records and 9 features using a continuous GitHub deployment workflow.
- Implemented user slider input widgets to funnel physiological vector attributes into a Scikit-Learn pipeline.
- Utilized a cached StandardScaler configuration to normalize inputs on the fly, eliminating structural model prediction bias and achieving 78.6% training and 77.2% testing accuracy.

Languages

- **English:** Fluent / Fließend (IELTS Academic Score: 7.0 / CEFR C1 Equivalent)
- **German:** A2/B1 (Active learner expanding vocabulary for quick integration into local teams)
- **Hindi / Gujarati:** Native proficiency

Hobbies & Interests

- **Competitive Gaming:** Passionate tactical gamer competing under the aliases PA1N and Painx09 across fast-paced, high-stakes environments including Valorant, Apex Legends, CS:GO, and Clash of Clans.
- **Physical Health & Exploration:** Maintain a balanced lifestyle outside of machine learning engineering through cooking, cycling, swimming, and travelling.